

CHAPTER 05

Forces in Nature

Three invisible forces shape everything in the universe — they hold you to the Earth, deliver the lightning during a storm, and stick your refrigerator magnet in place. Time to meet gravity, electricity, and magnetism.

⚡ Georgia Standard S8P5

WHAT YOU'LL LEARN

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| 01 Forces at a Distance | 06 Transferring Charges |
| 02 Gravity & Gravitational Fields | 07 Static Electricity & Fields |
| 03 Mass vs. Weight | 08 Magnetism & Magnetic Fields |
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5.1 Forces at a Distance

You drop your phone. Before it can even fall, you know what's going to happen — it heads straight for the floor. Nothing was touching it. No invisible string yanked it down. Yet a force **reached out** and pulled it. That's gravity. And it's just one of three forces in nature that can act on objects *without touching them*.

In Chapter 3 we learned that forces are pushes and pulls. Most of the forces you experience are **contact forces** — you have to touch something to push or pull it. But three special forces don't need contact at all:

- **Gravity** — pulls objects with mass toward each other
- **Electric force** — between objects with electric charge
- **Magnetic force** — between magnets and certain metals

These are **non-contact forces**. How do they work without touching? The answer is one of the coolest ideas in physics: **fields**.

What's a Field?

A **field** is an invisible area of influence around an object. Inside the field, other objects feel a force — even though nothing visible is connecting them. Think of it like an "aura" of force surrounding the object.

Three Kinds of Fields — One Big Idea

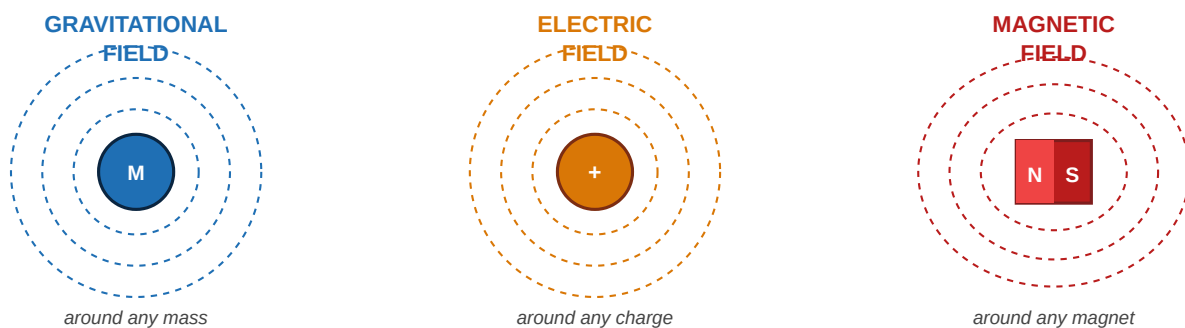


Figure 5.1 — Every gravity-producing, charged, or magnetic object has an invisible field around it. The field carries the force.

The stronger the field, the stronger the force. The farther you go from the source, the weaker the field gets.

! Did You Know?

The Voyager 1 spacecraft, launched in 1977, is now over 15 billion miles from Earth — and the Sun's gravity is *still* tugging on it. The Sun's gravitational field extends across our entire solar system and beyond. Fields can reach incredibly far, even though they get weaker with distance.

5.2 Gravity & Gravitational Fields

Gravity is a force of attraction between **any two objects that have mass**. Yes — any two objects. Not just Earth and the Moon. Right now, your textbook is being pulled toward you, and you're being pulled toward it. Both of you have mass, so gravity is acting between you.

Why don't you feel the pull from your textbook? Because gravity is an incredibly **weak** force. It only becomes noticeable when one of the objects is *massive* — like Earth (5.97×10^{24} kg of mass). That's why gravity dominates your day-to-day life: you live on a giant ball of mass.

Two Factors That Affect Gravity

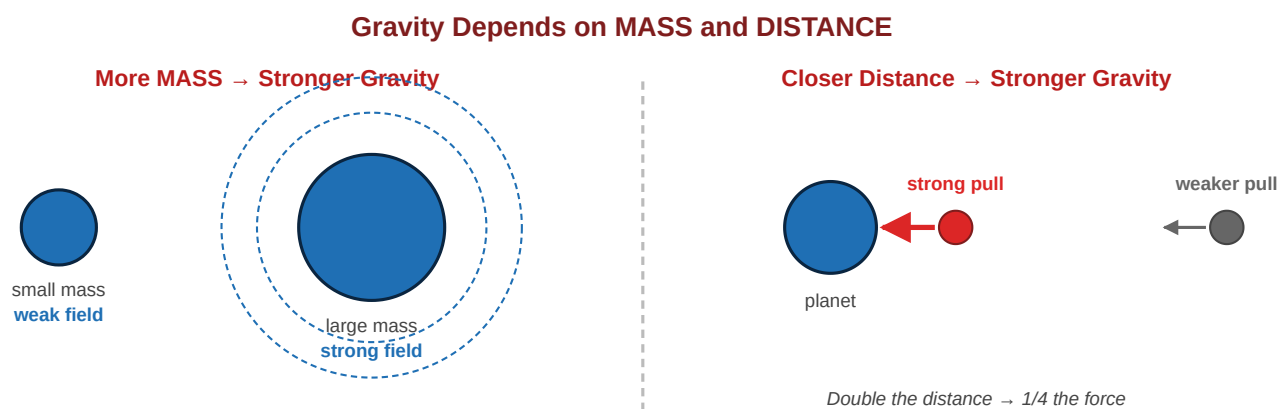


Figure 5.2 — Gravity gets stronger with more mass and weaker with more distance.

MORE MASS → MORE GRAVITY | MORE DISTANCE → LESS GRAVITY

! Did You Know?

The Sun pulls on Earth with a force of about **3.5×10^{22} newtons** — that's enough force to keep our entire planet curving in orbit instead of flying off into space. Jupiter has 318 times Earth's mass — if you stood on Jupiter, you'd weigh almost 2.5 times what you weigh here.

5.3 Mass vs. Weight

These two words get mixed up all the time, but in physics they mean very different things.

→ Mass

The **amount of matter** in an object. Mass **doesn't change** based on location. You have the same mass on Earth, on the Moon, in space, or anywhere else.
Measured in **kilograms (kg)**.

→ Weight

The **force of gravity** acting on an object's mass. Weight **changes** depending on where you are — different planets have different gravity.
Measured in **newtons (N)**.

Location	Gravity	Your Mass (50 kg student)	Your Weight
Earth	9.8 m/s ²	50 kg	490 N (~110 lbs)
Moon	1.6 m/s ²	50 kg	80 N (~18 lbs)
Mars	3.7 m/s ²	50 kg	185 N (~42 lbs)
Jupiter	24.8 m/s ²	50 kg	1,240 N (~280 lbs)
Outer space (no gravity nearby)	~0 m/s ²	50 kg	0 N (weightless)

X Common Mix-Up

When astronauts float around on the International Space Station, they're **NOT** weightless because there's no gravity. Earth's gravity is still pulling them strongly — about 90% as strong as on the surface. They appear weightless because the ISS is constantly **falling** around Earth (in orbit). Free-fall feels like weightlessness, but gravity is very much still there.

? Quick Check 5.3

1. If you have a mass of 40 kg on Earth, what is your mass on the Moon?
2. Why would your weight on Jupiter be greater than your weight on Earth?
3. Is "newtons" a unit of mass or weight? Explain.

5.4 Electric Charge

Now let's move to the second non-contact force: electricity. Everything starts with **electric charge** — a property of matter that causes it to feel a force in an electric field.

From Chapter 1, you already know two types of charged particles:

- **Protons** carry a **positive (+)** charge.
- **Electrons** carry a **negative (-)** charge.
- **Neutrons** are neutral (no charge).

In a normal atom, the number of protons equals the number of electrons — the positives and negatives cancel out, so the atom is **neutral**. But if electrons get added or stripped away, the atom (or object) becomes **charged**.

The Big Rule of Charges

Like Charges REPEL • Opposite Charges ATTRACT

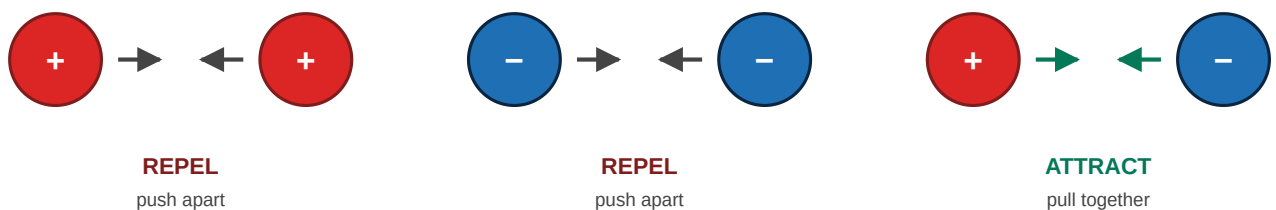


Figure 5.3 — Two charges of the same kind push each other away. Opposite charges pull each other together.

Like charges REPEL. Opposite charges ATTRACT.

5.5 Conductors & Insulators

Some materials let electric charges flow through them freely. Others basically block them. This depends on how tightly an atom holds onto its electrons.

How Charges Flow in Different Materials

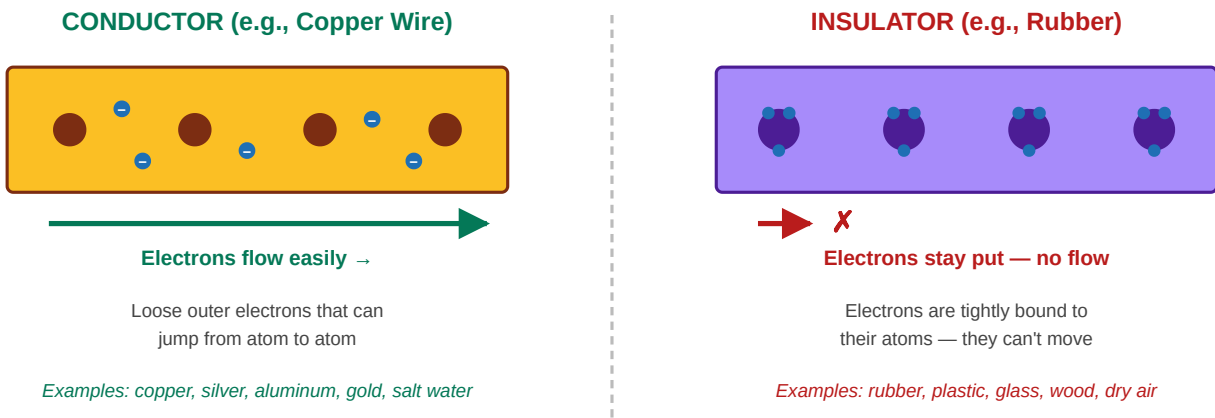


Figure 5.4 — Conductors let charges flow because their outer electrons are loose. Insulators trap their electrons in place.

Distribution of Charge

Here's something interesting: when a **conductor** gets charged, the charges **spread out evenly** across its surface — they want to get as far apart as possible (remember, like charges repel). When an **insulator** gets charged, the charges **stay right where you put them** — they can't move freely to spread out.

This is why power lines have a thick rubber or plastic coating around the wires. The metal carries the current (conductor), and the coating keeps the electricity from jumping out into your hand (insulator).

★ Real-World Connection

Why is the bird sitting on a power line not electrocuted? The bird's body would conduct electricity — but both feet are on the same wire, so there's no "circuit" for the current to flow through. If the bird touched the wire AND a grounded pole at the same time, electricity would flow through it. Same reason electricians wear thick rubber gloves: rubber is an insulator that breaks the path.

? Quick Check 5.5

1. Classify each as conductor or insulator: (a) a copper penny, (b) a wooden spoon, (c) a plastic phone case, (d) aluminum foil.
2. Why do electrical wires have a plastic coating around the metal inside?

5.6 Transferring Charges: Three Ways

So how do objects actually *become* charged? Charges don't appear out of nowhere — they always come from electrons moving from one place to another. There are **three different methods** of transferring charge between objects: friction, conduction, and induction.

Three Ways to Transfer Charge

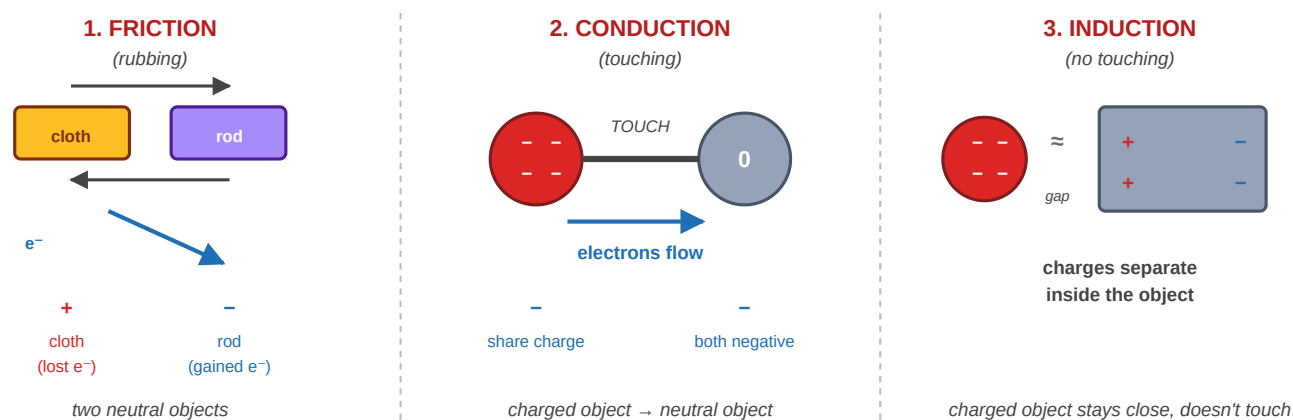


Figure 5.6 — Three methods of charge transfer. Each one moves electrons in a different way.

1. Charging by Friction (Rubbing)

When two different materials rub against each other, friction can **strip electrons** off one material and transfer them to the other. The material that *loses* electrons becomes positively charged. The material that *gains* electrons becomes negatively charged.

Some materials hold their electrons more tightly than others. When you rub them together, the one with the weaker grip on its electrons loses them. This is why specific combinations work so well — like wool and a balloon, or your shoes and a carpet.

Examples: Rubbing a balloon on your hair, shuffling across carpet in socks, pulling clothes out of a dryer (static cling).

2. Charging by Conduction (Touching)

When a **charged object touches a neutral object**, electrons flow directly from one to the other through the point of contact. The charge spreads out between them until both objects have the same charge — both negative if a negatively-charged object touched the neutral one, both positive if it was positively charged.

This only works well if at least one of the objects is a **conductor** (remember Section 5.5 — conductors let electrons move freely). If you touch a charged metal sphere to a neutral metal sphere, electrons rush across and the charge gets shared.

Examples: Touching a charged Van de Graaff generator and feeling your hair stand up, getting "zapped" when you grab a metal doorknob after walking on carpet.

3. Charging by Induction (Without Touching)

Here's the sneaky one. A charged object can affect a neutral object **without ever touching it**. When the charged object comes close, its electric field pushes or pulls on the electrons in the neutral object — causing the charges inside to **separate**.

For example, if you bring a negatively-charged balloon near a neutral wall, the electrons in the wall's surface get pushed away (because like charges repel). This leaves the surface of the wall with a slight **positive** charge — which then attracts the balloon. That's why a charged balloon will stick to a wall.

The neutral object hasn't actually gained or lost electrons overall — but its charges have rearranged so that one side is more positive and the other is more negative.

Examples: A charged balloon sticking to a wall, a comb picking up small bits of paper after running through your hair, lightning rods drawing charge from a thunderstorm before it strikes.

Method	How It Works	Do They Touch?	Final Charges
Friction	Rubbing strips electrons off one object	YES (rubbed together)	One becomes +, the other becomes –
Conduction	Electrons flow through direct contact	YES (touched)	Both end up with the SAME charge
Induction	Field pushes/pulls electrons within the neutral object	NO (just close)	Charges separate inside the neutral object

★ Real-World Connection

Touchscreens on your phone use **induction**. Your finger has a small electric charge on it, and when it gets close to the screen, it affects the electric field of the touch sensors underneath the glass — without actually transferring electrons. That's how the phone knows where you tapped. Try touching the screen with a pencil eraser — nothing happens, because rubber doesn't carry charge.

? Quick Check 5.6

1. You rub a plastic comb on a wool sweater and the comb becomes negatively charged. Which way did the electrons flow — comb to sweater, or sweater to comb?
2. Identify the method of charging: (a) a balloon sticks to a wall without rubbing, (b) sock fluff in the dryer, (c) touching a metal ball to a charged rod and shocking yourself.
3. In induction, does the neutral object actually gain or lose any electrons overall? Why does it still feel attracted to the charged object?

5.7 Static Electricity & Electric Fields

Static electricity is a **buildup of electric charge** on the surface of an object. It usually happens when two objects rub together — friction strips electrons off one object and dumps them onto the other.

The classic example: rub a balloon against your hair. The balloon picks up extra electrons from your hair, leaving the balloon negatively charged and your hair positively charged. Now your hair is attracted to the balloon (opposite charges), and individual hairs are repelled from each other (all positive), making them stand up.

Static Electricity: Balloon & Hair

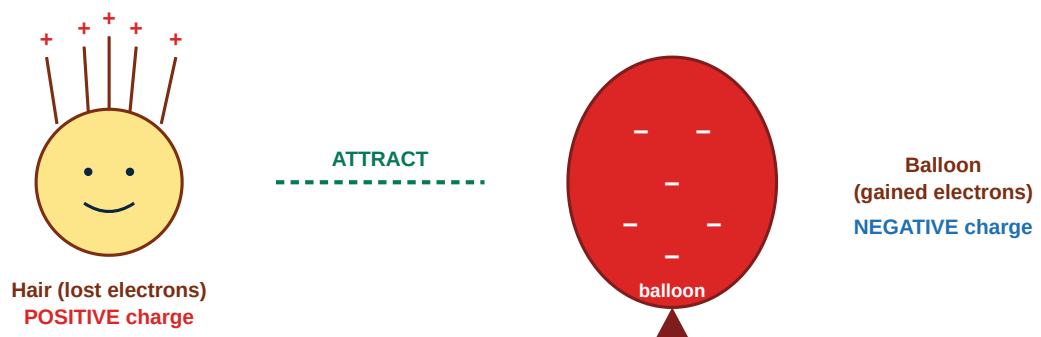


Figure 5.7 — Rubbing transfers electrons from hair to balloon. Hair becomes positive, balloon becomes negative — and they attract.

Lightning: The Ultimate Static Discharge

The same idea — at massive scale — causes lightning. Inside a thunderstorm, ice and water droplets collide and rub together. Electrons get stripped off, building up a huge negative charge at the bottom of the cloud and a positive charge at the top. When the charge difference gets large enough, electrons leap downward in a massive bolt — **lightning**. A single lightning bolt carries about 1 billion volts of electricity.

! Did You Know?

That little "zap" you feel when you walk across a carpet and touch a doorknob? That's a mini lightning strike. You picked up electrons from the carpet through friction, and when you got close to the metal doorknob (a conductor), those electrons jumped from you to the knob. The voltage is up to **5,000 volts** — but the current is so tiny, all you feel is a little snap.

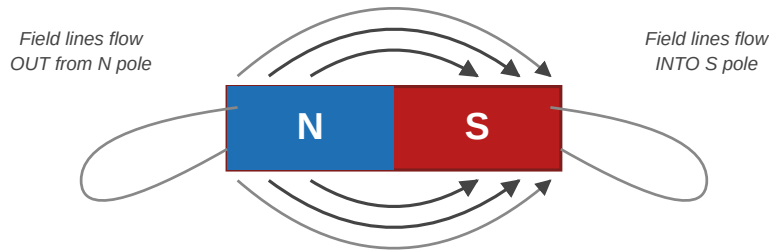
5.8 Magnetism & Magnetic Fields

A **magnet** is an object that produces a magnetic field and attracts iron, nickel, or cobalt. Every magnet has two ends called **poles**: a **North (N)** pole and a **South (S)** pole. Magnetic poles behave a lot like electric charges:

Like poles REPEL. Opposite poles ATTRACT.

Push two North poles together and they fight you. Bring North to South and they snap together. The magnetic field is strongest right at the poles and weaker in the middle of the magnet.

Magnetic Field Around a Bar Magnet



More field lines = stronger field (strongest at the poles)

Figure 5.8 — Magnetic field lines always loop from the North pole around to the South pole.

Earth Is a Giant Magnet

Our planet has a molten iron core that spins as Earth rotates. This generates a **massive magnetic field** around the entire planet. Earth essentially behaves like a giant bar magnet — which is why a compass works. The compass needle is a tiny magnet, and it aligns itself with Earth's magnetic field, always pointing toward magnetic north.

! Did You Know?

Earth's magnetic field is what protects us from deadly solar radiation. The Sun constantly blasts charged particles at us (the "solar wind"), but our magnetic field deflects most of them around the planet. Without it, life as we know it couldn't survive. The aurora (Northern Lights) is what we see when some of those particles do leak through near the poles.

5.9 Electromagnets

Here's where electricity and magnetism start to crash into each other. In 1820, a scientist named Hans Christian Ørsted discovered something wild: when an electric current flows through a wire, the wire **creates a magnetic field** around it!

If you coil that wire into many loops and wrap it around an iron core, you get an **electromagnet** — a magnet that you can turn on and off just by flipping a switch. Junkyard cranes use giant electromagnets to pick up cars: power on = pick up car, power off = drop car.

How an Electromagnet Works

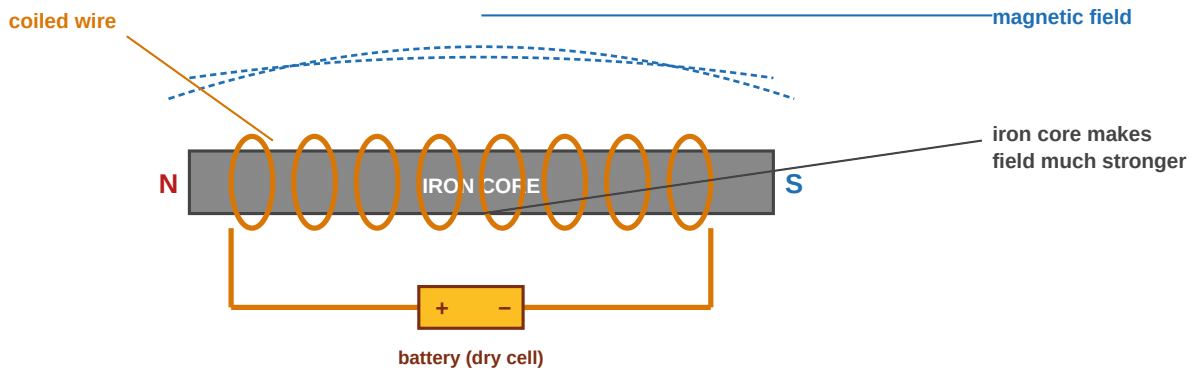


Figure 5.9 — Run current through a coiled wire wrapped around iron, and you get a temporary magnet you can switch on and off.

What Makes an Electromagnet Stronger?

The strength of an electromagnet depends on **four factors** you can control:

Factor	To Make It Stronger...
Number of wire turns (coils)	More loops of wire = stronger field
Amount of current	More current (more or bigger batteries) = stronger field
Size of the iron core	Larger iron core = stronger field
Material of the core	Iron, steel, and cobalt cores boost the field much more than empty air

★ Real-World Connection

Electromagnets are everywhere in your daily life. Doorbells, electric motors (including the one inside your fan), speakers, headphones, MRI machines, maglev trains, and the read/write heads in hard drives — all use electromagnets. Even your refrigerator has electromagnets in its compressor, keeping your snacks cold.

? Quick Check 5.9

1. You build an electromagnet but it's too weak to pick up a paperclip. Name three things you could change to make it stronger.
2. What's one BIG advantage an electromagnet has over a regular bar magnet?
3. If you removed the iron core from an electromagnet but kept the coil and battery, would it still work? Would it be as strong?

5.10 The Big Connection: Electricity ↔ Magnetism

What Ørsted discovered in 1820 — and what scientists spent the next century unpacking — is one of the most important ideas in all of physics:

Moving electric charges CREATE magnetic fields.
Moving magnetic fields CREATE electric currents.

Electricity and magnetism aren't really two separate things. They're two sides of the same coin. We even have a name for this combined force: **electromagnetism**.

And remember from Chapter 4 — what kind of waves did we say can travel through empty space? **Electromagnetic waves** (light, radio, microwaves, X-rays). Those waves exist because moving electric and magnetic fields can create each other endlessly, propagating across the universe at the speed of light. The light from a star a billion years away got to your eye because of this exact connection.

! Did You Know?

Almost every electric outlet in your house is powered by this idea in reverse. Power plants spin giant magnets near coils of wire. The moving magnetic field **creates** an electric current in the wire — which then travels through the power grid all the way to your phone charger. Whether it's a coal plant, a hydroelectric dam, a wind turbine, or a nuclear plant, they all use this same trick.

? Quick Check 5.10

1. What's the relationship between electricity and magnetism?
2. How does a power plant turn motion into electricity?
3. Why are light waves called "electromagnetic" waves?

★ CHAPTER 5 BIG IDEAS

- **Gravity, electric, and magnetic forces** can all act on objects without touching them. They're called **non-contact forces**.
- Every non-contact force works through an invisible **field** — the stronger the field, the stronger the force.
- **Gravity** depends on mass and distance: more mass = stronger pull; more distance = weaker pull.
- **Mass** (kg) stays the same everywhere. **Weight** (N) changes with gravity.
- **Like charges repel; opposite charges attract**. Same rule applies to magnetic poles.
- **Conductors** let charge flow freely (metals, salt water). On a conductor, charges spread out evenly.
- **Insulators** hold electrons tightly (rubber, plastic, glass). On an insulator, charges stay where you put them.
- **Static electricity** is built-up charge from friction. Lightning is a giant static discharge.
- Every magnet has **two poles** (N and S). The field is strongest at the poles. Earth is a giant magnet.
- An **electromagnet** is made by running current through a coiled wire — usually around an iron core. Strength depends on number of coils, current, and size/type of core.
- Electricity and magnetism are connected: **moving charges create magnetic fields; moving magnetic fields create electric currents**.

I Can... (Self-Check)

- ☐ I can argue with evidence that magnetic, gravitational, and electric fields exist between objects that exert forces on each other even when not in contact. (S8P5.a)
- ☐ I can describe the distribution of charge in conductors versus insulators. (S8P5.b)
- ☐ I can identify the factors that affect the strength of electric and magnetic forces — distance between objects, number of wire turns in an electromagnet, number/size of dry cells, and size of the iron core. (S8P5.c)